

Statistical modeling

Module 10

Marc-Olivier Boldi

Bachelor, GSEM

Autumn 2018

Logistic regression: Comparisons of models

Like with the F -test for regressions, logistic regression models can be compared using a statistical test.

However, there is no sum of squares because they are meaningless for binary (0/1) outcome. The sum of squares can be replaced by the log-likelihood value and the F -test by the so-called likelihood ratio test.

Remark: Except from the fact that the elements in the calculations are different because the underlying theory is slightly different, this method is used **exactly** in the same way as for regression. In particular, p -values are produced and models can be compared.

Logistic regression: Comparisons of models

Suppose we want to compare two models M_1 and M_0 . M_1 is the most complex model (it has more variables) and M_0 is nested within M_1 (i.e. it has the same variables except some of them). Then we have the following notations

- ▶ Models M_1 and M_0 , M_0 nested within M_1 ,
- ▶ Number of parameters in each models: $p_1 > p_0$,

To estimate the two models, the MLE is used, and the corresponding maximum log-likelihood are computed:

$$\ell_0(\hat{b}_0) = \arg \max_{M_0} \ell(b), \quad \ell_1(\hat{b}_1) = \arg \max_{M_1} \ell(b)$$

Logistic regression: Comparisons of models

Since M_0 is nested within M_1 , its maximum log-likelihood is smaller than the one of M_0 ,

$$\ell_0(\hat{b}_0) \leq \ell_1(\hat{b}_1).$$

The analogy for regression is that the sum of square of M_1 is always smaller than the sum of squares of M_0 .

As a consequence, the following difference is always non-negative

$$2 \left\{ \ell_1(\hat{b}_1) - \ell_0(\hat{b}_0) \right\} \geq 0.$$

The log-likelihood playing the same role as the sum of squares, the limit of this inequality,

$$\ell_1(\hat{b}_1) = \ell_0(\hat{b}_0),$$

corresponds to the two models are the same $M_0 = M_1$.

Logistic regression: Comparisons of models

In order to test the null hypothesis $H_0 : M_0 = M_1$, the following test plays the same role as the F -test for the regression.

The Likelihood Ratio Test (LRT) states that, if $H_0 : M_0 = M_1$ is true, then the following difference follows a $\chi^2_{p_1-p_0}$ distribution:

$$LRT = 2 \left\{ \ell_1(\hat{b}_1) - \ell_0(\hat{b}_0) \right\} \sim \chi^2_{p_1-p_0}$$

The test decision is thus to reject H_0 (at level α) if either

- ▶ $LRT > \chi^2_{p_1-p_0; 1-\alpha}$,
- ▶ $p\text{-value} = 1 - \chi^2_{p_1-p_0}(LRT) < \alpha$.

Logistic regression: Comparisons of models

Example: Is factor Age significantly (5%) associated with the response Renewed?

Like for the regression, the test can be done using function `anova`:

```
> mod1 <- glm(Renewed~Pages.Week + AvPages.Week + Age, data=Renew,
              family="binomial")
> mod0 <- glm(Renewed~Pages.Week + AvPages.Week, data=Renew,
              family="binomial")
> anova(mod0,mod1, test="LRT")
```

Analysis of Deviance Table

Model 1: Renewed ~ Pages.Week + AvPages.Week

Model 2: Renewed ~ Pages.Week + AvPages.Week + Age

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	57	61.203			
2	52	46.943	5	14.261	0.01404 *

Test: since $0.014 < 0.05$, H_0 is rejected. Thus Age is associated with Renewed.

Logistic regression: Comparisons of models

In details, R is computing the following elements

```
> mod1 <- glm(Renewed~Pages.Week + AvPages.Week + Age, data=Renew,
               family="binomial")
> mod0 <- glm(Renewed~Pages.Week + AvPages.Week, data=Renew,
               family="binomial")
> (ell0 <- logLik(mod0))
'log Lik.' -30.60175 (df=3)
> (ell1 <- logLik(mod1))
'log Lik.' -23.47138 (df=8)
> (LRT <- 2*(ell1 - ell0))
'log Lik.' 14.26075 (df=8)
> qchisq(0.95, df=5)
[1] 11.0705
```

That is, since $14.3 > 11.1$, H_0 is rejected (i.e. Age is significant at 5%).

Logistic regression: Comparisons of models

In the above example, the two models are

$$M_1 : \eta_1(x) = b_0 + b_1x_1 + b_2x_2 + b_{31}\delta_{g_1}(x_3) + b_{32}\delta_{g_2}(x_3) \\ + b_{33}\delta_{g_3}(x_3) + b_{34}\delta_{g_4}(x_3) + b_{35}\delta_{g_5}(x_3)$$

$$M_0 : \eta_0(x) = b_0 + b_1x_1 + b_2x_2.$$

The null hypothesis $H_0 : M_0 = M_1$ corresponds to

$$H_0 : b_{31} = b_{32} = b_{33} = b_{34} = b_{35}$$

that is,

" H_0 : all the coefficients associated to Age are 0"

or equivalently Age is not significant (i.e. there is no difference between age categories in terms of renewal)

Logistic regression: Comparisons of models

The number of parameters are respectively

- ▶ $p_0 = 3$

- ▶ $p_1 = 8$

And thus, indeed,

$$p_1 - p_0 = 5$$

which corresponds to the number of coefficients associated with Age and that are tested.

Logistic regression: model significance

In the case of a regression, the overall model significance was checked by an F -test. This may be done also for logistic regression by comparing this model (M_1) to the null model (M_0), i.e. containing only the intercept.

Unlike for linear regression, where the model significance is always checked by default at the moment of model fit, this is not automatic for logistic regression. It thus needs to be performed by the user by hand.

Logistic regression: model significance

Example: the full model is significant at 5%

$$M_1 : \eta_1(x) = b_0 + b_1x_1 + b_2x_2 + b_{31}\delta_{g_1}(x_3) + b_{32}\delta_{g_2}(x_3) \\ + b_{33}\delta_{g_3}(x_3) + b_{34}\delta_{g_4}(x_3) + b_{35}\delta_{g_5}(x_3)$$

$$M_0 : \eta_0(x) = b_0.$$

```
> mod1 <- glm(Renewed~Pages.Week + AvPages.Week + Age, data=Renew,  
              family="binomial")
```

```
> mod0 <- glm(Renewed~1, data=Renew, family="binomial")
```

```
> anova(mod0,mod1, test="LRT")
```

Analysis of Deviance Table

Model 1: Renewed ~ 1

Model 2: Renewed ~ Pages.Week + AvPages.Week + Age

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	59	80.761			
2	52	46.943	7	33.819	1.862e-05 ***

Logistic regression: individual tests on the coefficients

Continuing the analogy with linear regression, tests on the coefficients can be made. The underlying mathematical results are similar to the one for the regression. It states that, if b_j is the true coefficient value in the population, then

$$\frac{\hat{b}_j - b_j}{s_{\hat{b}_j}} \sim N(0, 1)$$

where

- ▶ $p + 1$ is the number of parameters in the model,
- ▶ " $\sim N(0, 1)$ " states for "is approximately distributed as a $N(0, 1)$ ",
- ▶ $s_{\hat{b}_j}$ is an estimate of the standard error of $\hat{b}_j$, which is a by-product of the fitting algorithm.

Logistic regression: tests on coefficients

Using the previous result, we have a test on one coefficient (**keeping all the other terms**), i.e. we test

$$H_0 : b_j = 0 \quad \text{versus} \quad H_1 : b_j \neq 0.$$

It uses the t -values

$$t_j = \hat{b}_j / s_{\hat{b}_j}, \quad j = 1, \dots, p$$

The test rejects H_0 at level α if either (equivalently)

- ▶ $|t_j| > z_{1-\alpha/2}$
- ▶ $p_j = 2 - 2 \times \text{pnorm}(t_j) < \alpha$

Logistic regression: tests on the coefficients

```
> summary(renew.mod)
Call:
glm(formula = Renewed ~ Pages.Week + AvPages.Week + Age, family = "binomial",
    data = Renew)
[...]
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.35667	1.24579	-1.892	0.0585 .
Pages.Week	0.08928	0.03564	2.505	0.0123 *
AvPages.Week	0.02350	0.02585	0.909	0.3634
Age25-29	0.68781	1.20624	0.570	0.5685
Age30-39	0.96324	1.27573	0.755	0.4502
Age40-49	-2.98274	1.37732	-2.166	0.0303 *
Age50-59	1.08553	1.16754	0.930	0.3525
Age60+	0.30892	1.26437	0.244	0.8070

```
[...]
```

Logistic regression: tests on the coefficients

It can be concluded that, at 5%,

- ▶ Factor `Pages.Week` is significant.
- ▶ Factor `AvPages.Week` is not significant **keeping all the other terms**.
- ▶ There is no significant difference between Age levels "18-24" and "25-29" (**keeping all the other terms**).
- ▶ There is no significant difference between Age levels "18-24" and "30-39" (**keeping all the other terms**).
- ▶ etc.

Logistic regression: confidence intervals

The theoretical t -value distribution allows to obtain an approximate confidence interval at $1 - \alpha$:

$$\begin{aligned} 1 - \alpha &= P \left(-z_{1-\alpha/2} \leq \frac{\hat{b}_j - b_j}{s_{\hat{b}_j}} \leq z_{1-\alpha/2} \right) \\ &= P \left(\hat{b}_j - z_{1-\alpha/2} s_{\hat{b}_j} \leq b_j \leq \hat{b}_j + z_{1-\alpha/2} s_{\hat{b}_j} \right). \end{aligned}$$

Therefore, an approximate confidence interval at $1 - \alpha$ is

$$\left[\hat{b}_j - z_{1-\alpha/2} s_{\hat{b}_j}, \hat{b}_j + z_{1-\alpha/2} s_{\hat{b}_j} \right].$$

The most common CI is at 95% ($\alpha = 5\%$), with $z_{0.975} = 1.96$, thus

$$\left[\hat{b}_j - 1.96 \times s_{\hat{b}_j}, \hat{b}_j + 1.96 \times s_{\hat{b}_j} \right].$$

Logistic regression: confidence intervals

Example: IC(95%) for b_1 (Pages.Week)

$$[0.089 \pm 1.96 \times 0.036] = [0.019, 0.159]$$

For all the others,

```
> confint.default(mod1)
```

	2.5 %	97.5 %
(Intercept)	-4.79838189	0.08504130
Pages.Week	0.01942094	0.15913334
AvPages.Week	-0.02717169	0.07417085
Age25-29	-1.67637430	3.05199536
Age30-39	-1.53715743	3.46362841
Age40-49	-5.68224201	-0.28323727
Age50-59	-1.20279550	3.37386306
Age60+	-2.16920353	2.78704580

Logistic regression: profile confidence intervals

Unlike for the regression, exact confidence intervals cannot be built for logistic regression. Slightly more precise confidence intervals than the normal one can be built using profile likelihood (outside the scope of this course). These intervals are asymmetric. When available, they should be preferred.

Example: By default R gives the profile confidence intervals

```
> confint(mod1)
Waiting for profiling to be done...
              2.5 %      97.5 %
(Intercept) -5.14449579 -0.15334930
Pages.Week   0.02606627  0.16734103
AvPages.Week -0.02582757  0.07933927
Age25-29     -1.71426098  3.16597735
Age30-39     -1.47409600  3.69456739
Age40-49     -6.31213265 -0.58343865
Age50-59     -1.14607725  3.56240074
Age60+       -2.20859149  2.88615495
```

Logistic regression: model diagnostics

Like for the regression, model assumptions have to be checked using residuals. The assumptions are

- ▶ The outcomes are distributed according to a Bernoulli law

$$P(Y_i = 1|x) = p(x), \quad P(Y_i = 0|x) = 1 - p(x).$$

- ▶ The observations are independent.

How can we define residuals for 0/1 data?

Logistic regression: model diagnostics

Two types of residuals are used

- Deviance residuals

$$d_i = s_i \sqrt{-2 \{y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)\}}$$

where $s_i = 2y_i - 1$ (i.e. $s_i = 1$ if $y_i = 1$, $s_i = -1$ if $y_i = 0$).

- Pearson residuals

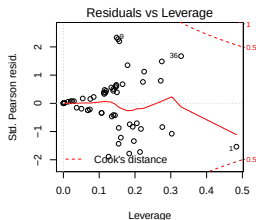
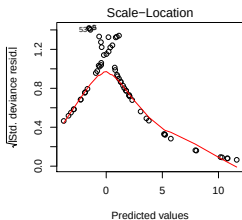
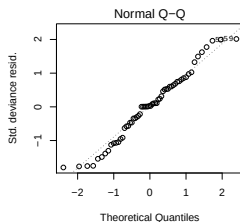
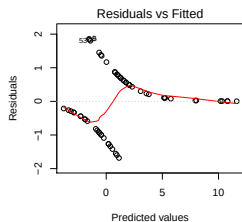
$$r_i = \frac{y_i - \hat{p}_i}{\sqrt{\hat{p}_i(1 - \hat{p}_i)}}.$$

Deviance residuals have slightly better theoretical properties than Pearson residuals, though they can both be used with the same aim (depending on the software you use).

Like for regression, if the model assumptions are validated by the data, the residuals should be distributed (approximately) as a $N(0, 1)$.

Logistic regression: model diagnostics

The 4 diagnostic plots are the same as for the regression.



Logistic regression: model diagnostics

- ▶ Top-left: deviance residuals (d_i) versus predicted values ($\hat{\eta}_i$).
- ▶ Top-right: QQ-plot normal versus d_i .
- ▶ Bottom-left: $\sqrt{|d_i|}$ versus $\hat{\eta}_i$.
- ▶ Bottom-right: r_i versus a leverage measure.

Logistic regression: model diagnostics

Top-Left: the upper points are $y_i = 1$. They are predicted as 1 when $\hat{\eta}_i$ is large (right side of the plot). Close to $x = 0$, the residuals increases, it is an uncertain zone. Same for lower points (symmetric). This is a classical pattern for logistic regression.

Top-right: QQ-plot should present the same pattern as in the regression case.

Bottom-left: results from the absolute value. The variance is higher when $\hat{\eta}_i$ is close to 0 (uncertain zone). It is very similar to the top-left plot.

Bottom-right: use the same as for the regression. It helps to detect influential points.